

MicroBlend — Network Plan

Topology

Internet → Modem → Router / Access Point (all Wi-Fi)

A single router provides both internet access and local LAN connectivity via Wi-Fi. No wired infrastructure — all devices connect wirelessly.

Devices

Device	Role	Notes
Server PC / Laptop	Hosts Django app + SQLite database	Runs during operating hours; shutdown at shift close. No cloud dependency for core operations.
POS Tablet	Waiter / Cashier station	Place orders, process payments, manage table sessions, resolve page requests.
Kitchen Display	Order queue for kitchen staff	Receives real-time order alerts and prep status updates via SSE.
Staff Phone	Mobile staff terminal	Handheld device for waiters and managers to monitor requests and tables.
Customer Phone	QR scan, menu browsing, ordering	Connects over venue Wi-Fi; no cellular data required.
Admin / Manager Laptop	Menu management, inventory, reports, cost simulations, audit logs	Administrative functions not needed on the POS floor.
External POS (cloud)	Third-party POS integration	Syncs orders and menu bidirectionally over the internet via API.

Data Flow

- **API + SSE** — All internal devices communicate with the Server via HTTP REST API and Server-Sent Events (SSE) for real-time push.
- **Sync Events** — Outbox pattern: the Server writes sync events to the database; a background worker pushes them to External POS. External POS also pushes order/menu updates back to the Server.
- **Internet dependency** — Only required for External POS sync. Core POS and table management functions work offline on the local Wi-Fi network.

Security Notes

- The router should have WPA2/3 encryption and a guest network isolated from the POS network.
- The Server PC should be configured to only accept LAN connections (no public-facing ports).
- External POS sync should use HTTPS with API keys.